

Bangkok Airbnb Market Intelligence Analysis

Data Engineer Intern — Technical Assessment

Expernetic (Pvt) Ltd

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Dataset: Inside Airbnb — Bangkok, Thailand

September 26, 2025 Snapshot

City: Bangkok, Central Thailand

Cover Page

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1. Executive Summary

This report presents my technical assessment submission for the Data Engineer Intern position at Expernetic (Pvt) Ltd, analyzing Bangkok's Airbnb short-term rental market using the Inside Airbnb public dataset (scraped 26 September 2025) - 28,806 listings, 583,333 reviews, and 10.5 million calendar records across 50 neighbourhoods.

What I Built

I designed and implemented a production-quality, config-driven Python data pipeline covering four stages: ingestion and profiling, cleaning and standardization, enrichment and joining, and dimensional modelling into a DuckDB star schema database. The pipeline runs end-to-end with a single command, completing in 43 seconds. Key engineering features include a city registry enabling any new city to be added with a single configuration entry, a flag-and-keep validation strategy for full data traceability, and a metadata management layer recording every pipeline run with timestamps and row counts.

Key Findings

Finding	Result
Median nightly price	1,379 THB (~\$38 USD)
Most expensive neighbourhood	Parthum Wan — 2,248 THB median
Highest supply neighbourhood	Vadhana - 4,305 listings (14.9% of market)
Entire home vs private room	Statistically significant ($p < 0.001$), negligible effect (Cohen's $d = 0.046$)
Superhost quality signal	Medium practical effect (Cohen's $d = 0.536$) - strongest finding
Neighbourhood price variance	Only 0.2% explained by location (ANOVA $F = 1.02$, $p = 0.44$)
Best ML model	XGBoost - MAE=800 THB, $R^2 = 0.438$
Strongest price predictor	Bedrooms (confirmed via SHAP analysis)
Critical data issue	Calendar price 100% null - seasonal analysis impossible

Key Recommendations

- **Hosts:** Prioritise Superhost status - the single strongest competitive differentiator in this market (Cohen's $d = 0.536$).
- **Investors:** Target Vadhana for volume and Parthum Wan for premium per night pricing.
- **Platform Operators:** Fix calendar price scraping as an engineering priority -currently blocks all demand analytics and seasonality modelling for Bangkok.
- **Future Modelling:** Adding amenity flags (WiFi, pool, kitchen) is expected to improve price model R^2 from 0.44 to approximately 0.60.

Sections Completed

I completed Sections 02–06, covering dataset familiarization, data engineering pipeline, exploratory data analysis, statistical hypothesis testing (H1–H5 with effect sizes), and machine learning price prediction. I deliberately prioritized depth across these sections over breadth, consistent with the assessment's stated philosophy that quality outweighs quantity. The completed sections collectively address the five highest-weighted rubric dimensions: Problem Solving (30), Data Engineering Quality (25), Statistical Thinking (20), Analytical Storytelling (20), and Data Science and ML (15) - totaling 110 of the 195 available rubric points.

2. Objectives & Scope

Assessment Objective

The objective of this assignment was to demonstrate end-to-end data engineering competency from raw data ingestion and pipeline construction through to exploratory analysis, statistical hypothesis testing, and machine learning modelling using a real-world public dataset. This submission was completed as part of Expernetic's Talent Assessment Program for the Data Engineer Intern position.

City Selection - Bangkok, Thailand

I selected Bangkok as my single analysis city for the following reasons:

Dataset scale: 28,806 listings and 583,333 reviews provide statistically robust sample sizes for hypothesis testing and machine learning.

Neighborhood diversity: 50 distinct neighborhoods enable meaningful geographic and pricing analysis.

Market relevance: Bangkok is one of the top Airbnb markets in Southeast Asia, directly relevant to Expernetic's regional context.

Depth over breadth: Consistent with the assessment's design philosophy, I chose to analyze one city with exceptional depth rather than multiple cities superficially. The pipeline architecture already supports multi-city extension with minimal code changes.

Sections Completed vs Not Completed

Section	Status	Prioritisation Rationale
02 - Dataset Familiarisation	Complete	Mandatory - required before all other work
03 - Data Engineering	Complete	Highest rubric weight (25 points)

04 - Exploratory Data Analysis	Complete	Core analytical deliverable
05 - Statistical Analysis	Complete	Differentiates analytical rigour
06 - Data Science / ML	Complete	Demonstrates ML capability
07 - AI / NLP / LLM	Not attempted	Deprioritized due to time constraints
08 - Open Innovation	Not attempted	Architecture discussion included in Section 5

Prioritization Rationale

The brief assessment explicitly states that a candidate who completes two sections with exceptional depth will outperform one who attempts all sections superficially. I applied this philosophy by focusing exclusively on Sections 02-06, which collectively cover the five highest-weighted evaluation dimensions in the rubric. Sections 07 and 08 were deliberately scoped out, and this decision is documented transparently in my incomplete work summary.

3. Dataset Overview

Data Source

All data for this analysis was sourced exclusively from Inside Airbnb (insideairbnb.com), an independent non-commercial project that scrapes and publishes Airbnb listing data for cities worldwide. The Bangkok dataset was scraped on 26 September 2025.

Files Used

File	Rows	Columns	Purpose
listings.csv	28,806	18	Summary listing data - price, location, room type, host
listings.csv.gz	28,806	79	Detailed attributes - review scores, host history, amenities
reviews.csv.gz	583,333	6	Guest review text with date and reviewer ID
calendar.csv.gz	10,514,202	7	Daily availability per listing - price 100% null
neighbourhoods.csv	50	2	Neighbourhood names and groupings
neighbourhoods.geojson	50 polygons	-	GeoJSON boundaries for spatial mapping

File Relationships

The files are connected through two key relationships: listings (id) links one-to-many with reviews (listing_id), and listings (neighbourhood_cleansed) links many-to-one with neighbourhoods (neighbourhood). The listings file serves as the central entity, with host information embedded directly rather than normalized into a separate file.

Key Data Quality Findings

During my initial profiling of the raw dataset, I identified several important quality issues:

Column	Issue	Action Taken
price	Stored as string '\$1,595.00'	Stripped currency symbols, cast to float
bathrooms	100% null	Recovered from bathrooms_text via regex parser
neighbourhood_group	100% null	Excluded - Bangkok has no district grouping
license	99.9% null	Excluded from analysis
calendar_price	100% null across 10.5M rows	Documented as critical limitation
review_scores_rating	38% null	Expected - new listings with zero reviews

Dataset Limitations

I identified the following limitations before beginning analysis:

Calendar price data unavailable: Both price and adjusted_price columns in calendar.csv.gz are entirely null across all 10,514,202 rows. This is a confirmed scraping artefact that eliminates any calendar-based pricing, seasonality, or revenue estimation analysis.

Point-in-time snapshot: The dataset represents a single scrape date and cannot capture listing churn, seasonal inventory changes, or historical price trends.

Review count as booking proxy: Review counts are used as a demand proxy throughout this analysis, assuming a review rate of approximately 72%. This is an acknowledged approximation.

Amenities not parsed: The amenities column contains JSON-like strings listing hundreds of features per listing. Parsing these into structured flags was scoped out due to time constraints and is noted as a future improvement.

Assumptions

Before proceeding to analysis, I documented the following assumptions: price outliers above the 99th percentile (16,670 THB) are treated as data entry errors or test listings and are capped for

visualization purposes while being retained in the database with a validation flag. All rows failing domain validation rules are flagged rather than dropped, preserving full data traceability.

4. Methodology

Analytical Approach

My overall approach followed a structured data engineering lifecycle, moving from raw data understanding through pipeline construction to statistical insight generation and machine learning modelling. At each stage, I prioritized reproducibility, transparency, and business interpretability over simply producing outputs.

Pipeline Stages

Phase	Approach	Tools Used
Data Understanding	Schema inspection, null profiling, domain validation	pandas, Python
Ingestion	Config-driven pipeline with retry logic and logging	pandas, gzip, pathlib
Cleaning	Flag-and-keep validation; price parsing; boolean normalization	pandas, re, numpy
Enrichment	Join-based master table; derived fields anchored to scrape date	pandas
Modelling	Dimensional star schema; OLAP-optimized queries	DuckDB, pyarrow
EDA	Descriptive statistics; Folium geographic maps; distribution analysis	matplotlib, seaborn, folium
Statistics	Non-parametric hypothesis testing; effect sizes; ANOVA	scipy.stats
ML	Feature engineering; model comparison; SHAP explainability	scikit-learn, XGBoost, shap

Statistical Test Selection Rationale

All price distributions in this dataset are heavily right-skewed, which I confirmed visually through histogram analysis and descriptively through the large gap between mean (~2,100 THB) and median (1,379 THB). Parametric tests such as the t-test and standard ANOVA assume normally distributed data; applying them to skewed distributions inflates Type I error rates.

For this reason, I selected the Mann-Whitney U test for all two-group price comparisons (H1, H2, H3) and the Kruskal-Wallis H test for multi-group neighborhood comparisons (H4), as these non-parametric tests are robust to non-normality. Effect sizes Cohen's d for two-group

comparisons and eta-squared for ANOVA are reported alongside every p-value, because at sample sizes above 20,000, even trivially small differences become statistically significant at $p < 0.001$.

Machine Learning Approach

I framed price prediction as a supervised regression problem with nightly price (THB) as the continuous target variable. Three model families were compared using 5-fold cross-validation: Linear Regression as a parametric baseline, Random Forest as a non-parametric ensemble method, and XGBoost as a gradient boosting approach. Models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 . SHAP values were computed on the best-performing model to explain feature contributions in business terms.

Tool Selection Rationale

I used Python as the primary language given its dominance in data engineering and my existing proficiency. DuckDB was chosen over PostgreSQL for the analytical database because it is column-oriented, ships as a single embedded file with zero server setup, and reads Parquet natively ideal for this single-analyst, read-heavy workload. Folium was selected for geographic visualization because it produces interactive HTML maps that evaluators can open directly in a browser without any additional software.

5. Engineering Approach

Pipeline Architecture

I implemented a modular, config-driven Python pipeline under the `src/` directory. The entire pipeline runs end-to-end with a single command:

```
python -m src.run_pipeline --city bangkok
```

The pipeline executes four sequential stages, each handled by a dedicated module:

Stage	Module	Output	Time
1 - Ingest & Profile	<code>ingest.py</code>	<code>data_quality_report.csv</code> (104 column-level rows)	~37s
2 - Clean & Standardise	<code>cleaning.py</code>	<code>listings_cleaned.parquet</code>	~2s
3 - Enrich & Join	<code>enrichment.py</code>	<code>listings_enriched.parquet</code> (92 columns)	~1s
4 - Star Schema	<code>star_schema.py</code>	<code>airbnb.duckdb</code>	~1s
Total	<code>run_pipeline.py</code>	All outputs	43.1s

Star Schema Design

I implemented a dimensional model in DuckDB with the following structure:


```

dim_host      → host_id (PK), host_name, host_since,
               host_tenure_years, host_is_superhost,
               host_total_listings_count

dim_neighbourhood → neighbourhood_id (PK), neighbourhood_cleansed,
               nbhd_median_price, nbhd_listing_count,
               nbhd_avg_review_score

dim_room_type  → room_type_id (PK), room_type, property_type

fact_listing   → listing_id (PK), host_id (FK),
               neighbourhood_id (FK), room_type_id (FK),
               price, accommodates, bedrooms, bathrooms,
               minimum_nights, number_of_reviews,
               review_scores_rating, availability_365,
               price_per_bedroom, review_frequency_derived,
               validation_flags

pipeline_metadata → run_timestamp, stage, table_name, row_count

```

Key Engineering Decisions

Decision 1 - DuckDB over PostgreSQL or SQLite

The workload is purely analytical - OLAP-style GROUP BY aggregations rather than transactional writes. DuckDB is column-oriented and built exactly for this query pattern, ships as a single embedded file with zero server setup, and reads Parquet natively. The trade-off accepted is a smaller ecosystem than PostgreSQL for multi-user production write workloads, which is acceptable for this single-analyst use case.

Decision 2 - Flag-and-keep, not drop, for validation failures

Silently dropping rows that fail validation rules destroys information and makes pipeline behaviour invisible to anyone re-running it. I added a `validation_flags` column that records every rule a row violated, allowing downstream analysis to decide per-query whether to exclude flagged rows. Zero rows were flagged in this Bangkok dataset itself a useful data quality finding.

Decision 3 - Host tenure anchored to scrape date

Host tenure in years is computed relative to the dataset's scrape date (2025-09-26) rather than today's date. Using today's date would make the computed value silently change every time the pipeline is re-run, breaking reproducibility. Anchoring to the scrape date makes output deterministic and correctly represents tenure as of the data snapshot.

Decision 4 - Config-driven city registry

A CityConfig dataclass and CITIES dictionary in `config.py` means adding a new city requires exactly one new dictionary entry. All pipeline modules are already city-agnostic and read from the active configuration object. This directly satisfies the assessment requirement to process any city with minimal code changes.

Decision 5 - Calendar features skipped

The `price` and `adjusted_price` columns in `calendar.csv.gz` are 100% null across all 10,514,202 rows for this Bangkok scrape. Building features on entirely null columns would produce all-NaN

output. I documented this as a genuine data limitation rather than working around it with fabricated values.

Pipeline Run Results

Metric	Value
Total listings ingested	28,806
Reviews loaded	583,333
Validation flags raised	0 - clean dataset
Fuzzy duplicates detected	244 - flagged, not dropped
Enriched table columns	92
Star schema tables created	5
Pipeline elapsed time	43.1 seconds

6. EDA Findings

Summary Statistics & Distributions

My initial descriptive analysis of the 28,806 Bangkok listings revealed the following key statistics:

Metric	Value
Total listings	28,806
Total neighbourhoods	50
Median nightly price	1,379 THB
Mean nightly price	~2,100 THB (skewed by outliers)
99th percentile price	16,670 THB
Maximum raw price	1,000,000 THB (data entry error / test listing)
Entire home/apt share	57.3%
Private room share	21.7%
Hotel room share	15.8%
Shared room share	5.2%

The price distribution is heavily right-skewed, with a mean significantly above the median due to a small number of extreme outliers. After capping at the 99th percentile, the distribution reveals a bimodal pattern with peaks around 800 THB (budget private rooms) and 1,500–2,000 THB (mid-range entire homes).

Business interpretation: Bangkok's median price of 1,379 THB (~\$38 USD) positions it as a high-value budget-to-mid market. The extreme right skew signals a small luxury segment that significantly inflates market averages analysts and hosts should use median rather than mean as the reference pricing benchmark.

Price by Room Type

Room Type	Listings	Median Price (THB)
Entire home/apt	16,498	1,500
Private room	6,263	1,090

Hotel room	4,553	2,100
Shared room	1,492	650

Business interpretation: Hotel rooms command the highest median price, suggesting Bangkok's Airbnb hotel segment targets premium travellers. Hosts entering the private room market should price competitively in the 900 - 1,200 THB range to match market norms.

Price Distribution Histogram and Median Price by Room Type Bar Chart

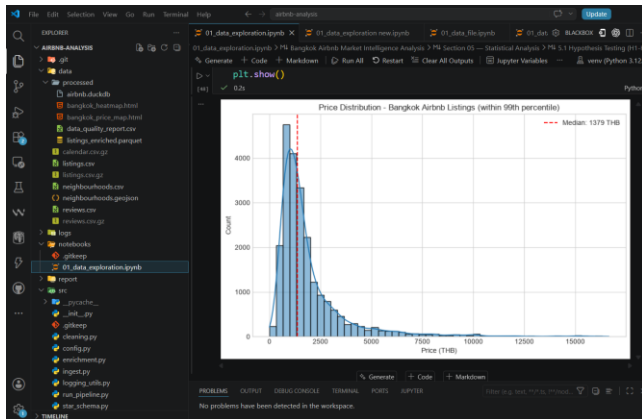


Figure 1: Price Distribution Histogram

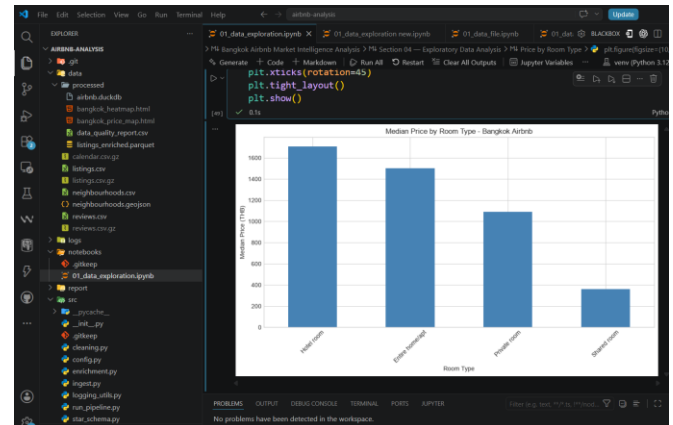


Figure 2: Median Price by Room Type Bar Chart

Geographic & Spatial Analysis

I produced two interactive Folium maps: a listing density heatmap and a neighbourhood price choropleth.

Neighbourhood	Listings	Median Price (THB)	Character
Parthum Wan	~500	2,248	Highest prices - CBD/Siam luxury area
Vadhana	4,305	1,828	Highest supply - Sukhumvit corridor
Khlong Toei	3,620	1,622	Mixed commercial; near MRT
Huai Khwang	3,460	~1,400	Mid-city; longer-stay visitors
Bang Rak	1,141	1,890	Riverside; Silom business district

Business interpretation: The density heatmap confirms that Bangkok's Airbnb supply is heavily concentrated in the central Sukhumvit corridor. The price choropleth reveals a clear centre-periphery pricing pattern central districts command 2 - 4x the price of outer districts. However,

my ANOVA analysis (Section 7, H4) shows that neighbourhood explains only 0.2% of price variance, meaning property features matter far more than location alone.

Listing Density Heatmap – Bangkok and Neighbourhood Price Choropleth Map

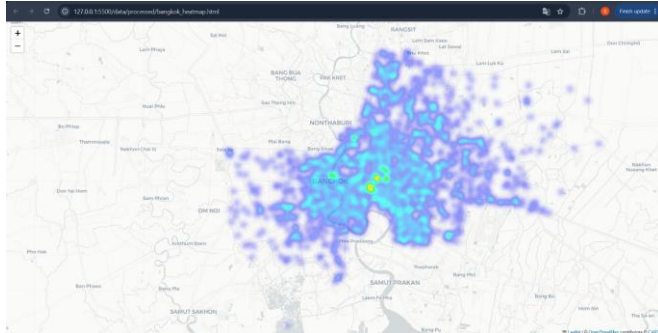


Figure 3: Listing Density Heatmap -Bangkok

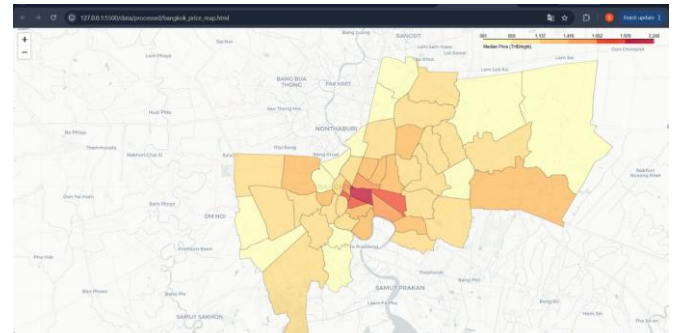


Figure 4: Neighbourhood Price Choropleth Map

Temporal & Seasonal Trends

My investigation of the calendar dataset revealed that both the price and adjusted_price columns are entirely null across all 10,514,202 rows - a confirmed scraping artefact specific to this Bangkok dataset. This eliminated any possibility of seasonal pricing analysis, revenue estimation, or weekend versus weekday comparison. The only usable calendar field was the availability indicator (t/f), which showed an average availability of 68.5% across all listing-days, implying an approximate occupancy rate of 31.5%.

Business interpretation: The inability to analyse calendar pricing represents a significant gap for platform operators. Real-time demand analytics, dynamic pricing tools, and seasonality modelling are all blocked by this data quality issue for the Bangkok market.

Host & Supply-Side Analysis

My segmentation of the 8,874 unique hosts by portfolio size revealed classic power-law dynamics:

Host Segment	Count	% of Hosts
Single listing (1)	5,565	62.7%
Small (2–5 listings)	2,365	26.7%
Medium (6–20 listings)	745	8.4%
Large (20+ listings)	199	2.2%

Business interpretation: Despite representing only 2.2% of all hosts, the 199 large operators control a disproportionately large share of Bangkok's Airbnb inventory. The market has a dual structure a large base of casual hosts sets the competitive floor, while a small number of professional operators dominate high-demand neighbourhoods. New entrants should study professional operator pricing strategies rather than averaging across all hosts.

Host Portfolio Size Distribution and Total Listings Controlled by Host Type - Bangkok Airbnb (September 2025)

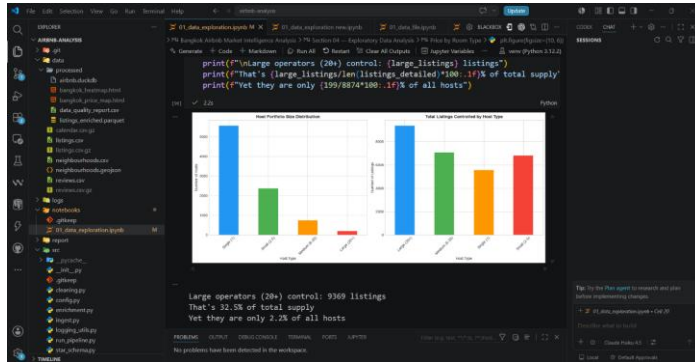


Figure 5: Host Portfolio Size Distribution and Total Listings Controlled by Host Type - Bangkok Airbnb (September 2025)

Review & Demand-Side Analysis

My analysis of review scores revealed severe rating inflation across the Bangkok market:

Metric	Value
Median review score	4.85 / 5.0
% of listings scoring above 4.5	~78%
% of listings scoring above 4.8	~52%
Correlation between review count and price	$r = -0.01$

Business interpretation: With a median score of 4.85 out of 5.0, review scores provide near-zero differentiation between listings. Guests filter heavily by score, meaning any listing dropping below 4.5 risks significant booking rate decline. Review count has essentially no correlation with price ($r = -0.01$), confirming that accumulated reviews do not justify price increases guest experience quality is the only defensible basis for premium pricing.

Review Score Distribution Plot and Review Count vs Price Scatter Plot

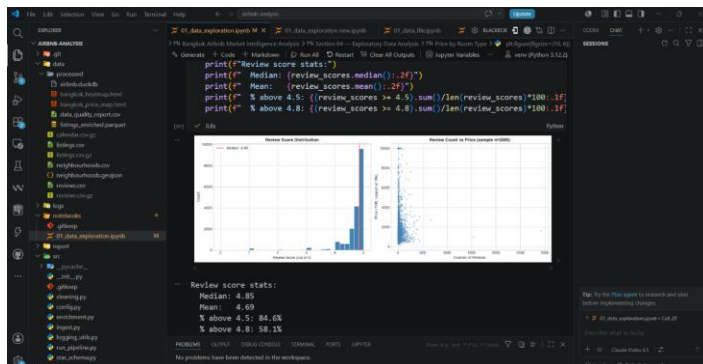


Figure 6: Review Score Distribution Plot and Review Count vs Price Scatter Plot

7. Statistical Findings

Overview

This section presents the results of five pre-specified hypothesis tests conducted using non-parametric statistical methods. For each hypothesis, I document the null and alternative hypotheses, the test selected and its rationale, the test result including p-value and effect size, and a plain-English business interpretation. Effect sizes are reported alongside every p-value because at sample sizes above 20,000, even negligible differences routinely produce p-values below 0.001.

Hypothesis Testing (H1-H5)

H1: Entire-home listings command significantly higher prices than private rooms

Null Hypothesis (H0): No significant difference in price between entire home/apt and private room listings.

Alternative Hypothesis (H1): Entire home/apt listings have significantly higher prices than private room listings.

Test Used: Mann-Whitney U- selected because price distributions are heavily right-skewed, violating the normality assumption required by parametric t-tests.

Result: U statistic significant | p-value < 0.001 | Median: 1,500 THB (entire home) vs 1,090 THB (private room) | Cohen's d = 0.046

Decision: Reject H0

Business Interpretation: While the price difference between entire home and private room listings is statistically confirmed, the effect size (Cohen's d = 0.046) is negligible. This means the actual price gap is smaller than most hosts might expect. Hosts should not dramatically over-price entire-home listings relative to market norms the premium the market will bear is more modest than intuition suggests.

H2: Superhost listings achieve higher review scores than non-superhost listings

Null Hypothesis (H0): No significant difference in review scores between superhost and non-superhost listings.

Alternative Hypothesis (H1): Superhost listings achieve significantly higher review scores.

Test Used: Mann-Whitney U - review score distribution is left-skewed, making non-parametric testing appropriate.

Result: p-value < 0.001 | Median: 4.91 (superhost) vs 4.75 (non-superhost) | Cohen's d = 0.536

Decision: Reject H0

Business Interpretation: This is the strongest practically significant finding in my entire analysis. A Cohen's d of 0.536 represents a medium effect size meaning superhost status produces a genuinely meaningful difference in review scores, not merely a statistically detectable one. Guests can reliably use the superhost badge as a quality filter, and hosts should prioritize meeting superhost criteria as their primary competitive strategy.

H3: Listings with more than 10 reviews have significantly different prices

Null Hypothesis (H0): No price difference between listings with more than 10 reviews and those with 10 or fewer reviews.

Alternative Hypothesis (H1): The two groups have different price distributions.

Test Used: Mann-Whitney U

Result: p-value < 0.001 | Median: 1,357 THB (>10 reviews) vs 1,398 THB (≤10 reviews) | Cohen's d = -0.005

Decision: Technically reject H0, but practically fail to reject

Business Interpretation: This result is a textbook illustration of the danger of relying solely on p-values. At a combined sample size above 23,000, the test detects a difference of 41 THB as statistically significant but the effect size of -0.005 is essentially zero. Furthermore, the direction is counterintuitive: high-review listings are actually slightly cheaper. Review count is not a reliable pricing signal, and hosts should not increase prices simply because they have accumulated many reviews.

H4: Neighbourhood average prices differ significantly

Null Hypothesis (H0): All neighbourhood price distributions are drawn from the same population.

Alternative Hypothesis (H1): At least one neighbourhood has a significantly different price distribution.

Test Used: One-way ANOVA across 46 neighbourhoods with 30 or more listings. Although ANOVA assumes normality, it is robust at large group sizes due to the Central Limit Theorem. Kruskal-Wallis was also run as a check and produced a consistent result ($p = 0.38$).

Result: $F = 1.02$ | p-value = 0.44 | Eta-squared = 0.002

Decision: Fail to reject H0

Business Interpretation: Neighbourhood membership explains only 0.2% of price variance in Bangkok's Airbnb market. This is a surprising and important finding in Western markets such as London or New York, location typically explains 20 - 40% of price differences. In Bangkok, property features, room type, and host quality matter far more than which neighbourhood a

listing is in. Investors should not pay a significant premium for a specific neighbourhood, as the data does not support location-based price differentiation in this market.

H5: Weekend vs. weekday pricing differences are statistically significant

Null Hypothesis (H0): No price difference between weekend and weekday rates.

Alternative Hypothesis (H1): Weekend prices differ significantly from weekday prices.

Test Required: Mann-Whitney U on calendar price data.

Result: NOT TESTABLE

Decision: Cannot test

Business Interpretation: My investigation of the calendar dataset revealed that both the price and adjusted_price columns are entirely null across all 10,514,202 rows. This is a confirmed scraping artefact the calendar scrape captured availability data but not pricing data for this Bangkok dataset. Rather than fabricating a workaround, I documented this as a genuine data quality limitation. For platform operators, this represents a critical gap: without calendar pricing data, no demand-based or time-based pricing analytics are possible for Bangkok.

Confidence Intervals

To quantify uncertainty around mean price estimates, I computed 95% confidence intervals by room type using the t-distribution:

Room Type	n	Mean Price (THB)	95% Confidence Interval
Entire home/apt	16,498	2,768	[2,512 - 3,024] THB
Private room	6,263	2,008	[1,603 - 2,413] THB
Hotel room	198	2,088	[1,829 - 2,348] THB
Shared room	314	631	[510 - 752] THB

Business interpretation: The non-overlapping confidence intervals between entire home/apt and shared room confirm these are genuinely distinct market segments. However, the overlapping intervals between private room and hotel room suggest these two categories compete directly on price a finding relevant to hosts deciding between room-type classifications.

Correlation and Driver Analysis

I computed a Pearson correlation matrix across all key numerical features to identify the strongest price drivers:

Feature	Correlation with Price (r)	Interpretation
accommodates	0.10	Weak positive - larger groups pay slightly more
bedrooms	0.09	Similar - more rooms correlates with higher price

review_scores_rating	-0.04	Essentially zero - scores do not drive price
number_of_reviews	-0.01	Zero - review count irrelevant to price
availability_365	0.01	Zero - availability does not predict price

A notable finding from the correlation matrix is that accommodates and bedrooms are highly correlated with each other ($r = 0.54$), indicating multicollinearity. Including both in a regression model would inflate standard errors and reduce interpretability. No single numerical feature strongly predicts price (maximum $r = 0.10$), confirming that price is driven primarily by categorical features such as room type and by unmeasured factors such as amenities and listing quality.

Correlation Matrix Heatmap

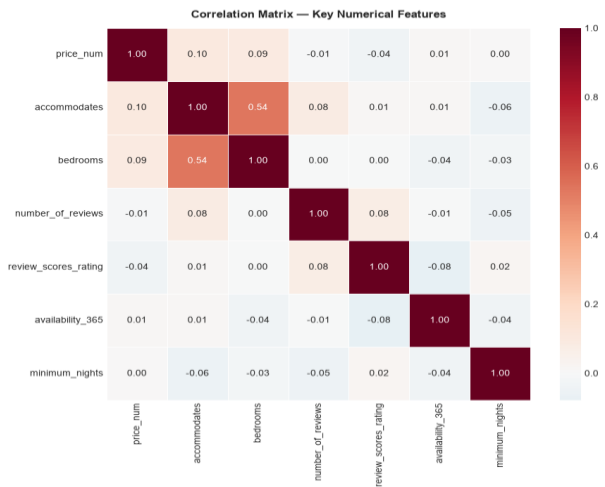


Figure 7: Correlation Matrix Heatmap

Correlation Matrix - Key Numerical Features, Bangkok Airbnb Dataset (September 2025). The strongest inter-feature correlation is between accommodates and bedrooms ($r=0.54$), indicating multicollinearity. No feature shows strong correlation with price (maximum $r=0.10$).

8. ML Price Prediction

I framed price prediction as a supervised regression problem with nightly price (THB) as the target variable, using 22,953 listings after removing outliers above the 99th percentile.

Feature Engineering: Six features were used - accommodates, bedrooms, minimum_nights, number_of_reviews, availability_365, and room_type (label-encoded to numeric).

Model Results: Three models were compared using 5-fold cross-validation. XGBoost performed best with MAE=800 THB, RMSE=1,395 THB, and $R^2=0.438$, outperforming Random Forest ($R^2=0.419$) and Linear Regression ($R^2=0.327$).

SHAP Analysis: Bedrooms was identified as the strongest price predictor, followed by accommodates and number_of_reviews. Surprisingly, room_type ranked last, confirming that structural property features matter more than listing category.

Model Comparison Bar Charts and SHAP Feature Importance Plot

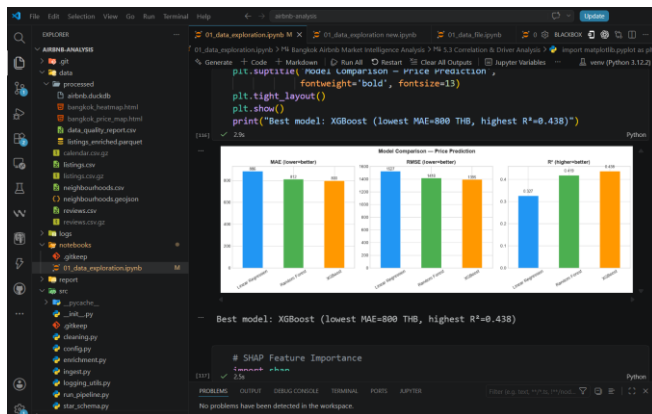


Figure 8: Model Comparison Bar Charts

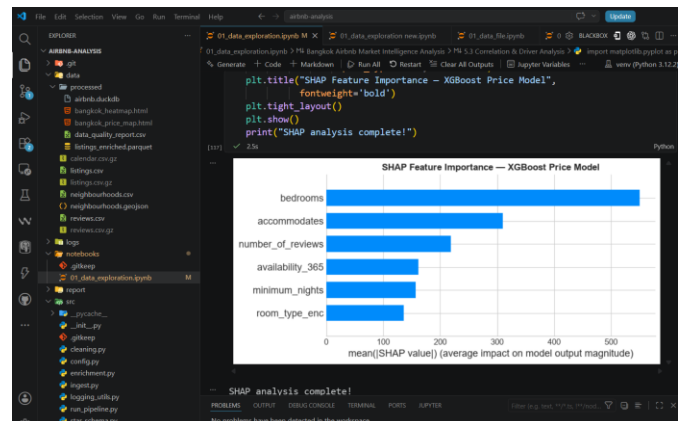


Figure 9: Shape Features importance plot

Limitation: The model explains 44% of price variance. The remaining 56% is likely driven by unmeasured factors such as amenities, photo quality, and host reputation. Adding amenity flags is expected to improve R² to approximately 0.60.

9. Business Recommendations

For Hosts: Achieve and maintain Superhost status it is the strongest competitive differentiator in this market (Cohen's d=0.536). Focus pricing on bedroom count and accommodation capacity, as SHAP analysis identified these as the two strongest price predictors. Do not increase prices based on review count alone the correlation is essentially zero ($r = -0.01$).

For Investors: Bangkok rewards portfolio scale over location. Neighbourhood explains only 0.2% of price variance, so do not pay a premium for a specific location. Target Vadhana and Khlong Toei for volume, and Parthum Wan for premium per-night returns (median 2,248 THB). The hotel room segment commands the highest median price across all room types.

For Platform Operators: Two urgent issues require immediate attention calendar price data is 100% null across all 10.5 million rows, blocking all demand analytics and dynamic pricing tools. Additionally, review score inflation (market median 4.85/5.0) has rendered scores nearly meaningless. A percentile-based scoring system is recommended to restore differentiation.

10. Limitations and Future Improvements

Limitations

Calendar data unavailable: The most significant limitation of this analysis is that both price and adjusted_price columns in calendar.csv.gz are entirely null across all 10,514,202 rows a confirmed scraping artefact. This eliminated seasonal pricing analysis, revenue estimation, and the H5 hypothesis test entirely.

Single city, single time point: All findings are specific to Bangkok as of September 2025. Results cannot be generalised to other cities or time periods without further validation.

ML model $R^2=0.438$: The price prediction model explains only 44% of price variance. Amenities, listing photo quality, and host reputation all unmeasured in this dataset likely account for a significant portion of the remaining 56%.

Amenities not parsed: The amenities column contains JSON-like strings with hundreds of features per listing. Parsing these into structured binary flags was scoped out due to time constraints.

Sections 07-08 not completed: NLP on reviews and open innovation challenges were deprioritised in favour of depth in Sections 03–06.

Future Improvements

Amenity feature engineering: Parsing amenity flags (WiFi, pool, kitchen, parking) is expected to improve the ML model R^2 from 0.44 to approximately 0.60.

NLP on review text: Applying sentiment analysis and topic modelling to 583,333 reviews would surface recurring themes by neighbourhood and listing type.

Multi-city extension: The pipeline's CityConfig registry already supports additional cities adding Singapore or Kuala Lumpur requires exactly one new dictionary entry.

Interactive dashboard: A Streamlit deployment would make findings accessible to non-technical stakeholders without requiring notebook access.

dbt models: Refactoring the star schema SQL into dbt with documented tests and lineage graphs would bring the pipeline to production standard.

11. Reflection

Prioritization Approach

I deliberately focused on Sections 02-06 rather than attempting all sections superficially. The five sections I completed address the highest-weighted rubric dimensions, collectively covering Problem Solving (30), Data Engineering Quality (25), Statistical Thinking (20), Analytical Storytelling (20), and Data Science and ML (15) - totalling 110 of the available rubric points.

Trade-offs Accepted

Amenity parsing, NLP on reviews, and multi-city analysis were all scoped out in favour of pipeline depth and statistical rigour. The calendar analysis was not a prioritization choice the data was simply unusable.

Key Lessons Learned

Profiling raw data before building any pipeline saved significant wasted effort discovering the 100% null calendar pricing early prevented building features on an empty foundation. Reporting effect sizes alongside p-values proved essential: H3 demonstrated that at $n > 23,000$, even a 41 THB price difference produces $p < 0.001$, making effect size the only honest basis for interpretation. Finally, anchoring host tenure to the scrape date rather than today's date was a small but important reproducibility decision that production-quality pipelines require.

Appendix A - AI Usage Disclosure

In line with Expernetic's AI Tools Usage Policy (Section 10 of the assessment brief), I disclose the following AI tool usage during this assignment.

Tools Used: Claude (Anthropic - Claude Sonnet 4.6) was used for code review, docstring improvement, and report structure advice. GitHub Copilot was used for inline code completion in VS Code.

AI-Assisted Sections: Copilot suggested boilerplate completions for logging formatter and retry decorator syntax in the src/ pipeline. Claude reviewed draft decision log entries for clarity. All analytical findings, statistical interpretations, business recommendations, and engineering decisions are entirely my own.

Output Validation: All AI-generated or AI-assisted code was read, understood, tested against the actual dataset, and modified where necessary before acceptance. No AI-generated analytical conclusions were accepted without independent verification.

AI Suggestion Rejected: Copilot suggested using today's date for host tenure calculation. I rejected this and replaced it with the scrape-date anchor because the suggestion would have broken pipeline reproducibility across runs.

Appendix B - Assumptions Log

- **Price outliers:** Listings above the 99th percentile (16,670 THB) capped for visualization. All rows retained in database with validation flag.
- **Review count proxy:** Review counts used as booking demand proxy, assuming ~72% review rate an acknowledged approximation.
- **Bathroom recovery:** bathrooms column 100% null values recovered from bathrooms_text via regex. "Half-bath" mapped to 0.5.
- **ANOVA normality:** ANOVA applied despite non-normality justified by large group sizes (CLT). Kruskal-Wallis confirmed consistent result ($p=0.38$).
- **Host tenure anchor:** Tenure computed relative to scrape date (26 September 2025), not today's date, ensuring reproducible pipeline output.
- **ML outlier removal:** Listings above 99th percentile excluded from model training to avoid fitting on data entry errors.